

# Long-context DNA models do not use the allele: an audit of DNALongBench’s eQTL task

Karandeep Nanda  
University of Colorado Boulder

## Abstract

eQTL prediction (eQTLP) is the only variant task in DNALongBench (Cheng et al., 2025), a standard benchmark for long-context DNA models. Ranking variants by distance to the transcription start site (TSS), with no training, reaches a mean test AUROC of 0.751, beating the expert model (0.681) and every foundation model (FM; 0.51–0.57). The test sets hold 11–34 positives per tissue, and every test gene also appears in training. We rebuild the task with distance-matched negatives and chromosome holdout, where hand-crafted local sequence features set a floor of 0.599. Scoring each pair again with the alternate allele replaced by the reference shows that no model uses the allele: for a CNN trained end to end and for frozen HyenaDNA and Caduceus, the swap changes AUROC by at most 0.0017, and all three fall below the floor. Fine-tuning HyenaDNA end to end over 3 folds does not change this: accuracy improves to the frozen probe’s level while the allele gap stays at  $|\Delta| \leq 0.00016$ . A nonlinear probe on the allele difference alone recovers only a weak trace (0.52–0.55). A single-nucleotide change moves the pooled FM embedding by about  $10^{-4}$  of its norm, and the released FM eQTL code slices every input to length zero. On 652 ClinVar pathogenic regulatory SNVs matched to benign SNVs, neither the embedding shift nor the transferred probe separates the classes (AUROC 0.49–0.51). We release the rebuilt task, the ablation harness, and the matched ClinVar set at <https://github.com/karan-nanda/Extending-DNALongBench>.

## 1 Introduction

Long-context DNA models (Nguyen et al., 2023; Schiff et al., 2024; Brixi et al., 2026) are motivated by distal regulation, meaning variants that act on genes tens to hundreds of kilobases away. They inherit their architectures from long-convolution and state-space sequence models (Poli et al., 2023; Gu and Dao, 2023), and are benchmarked against expert models such as Enformer (Avsec et al., 2021), Akita (Fudenberg et al., 2020) and Puffin (Dudnyk et al., 2024). DNALongBench (Cheng et al., 2025) evaluates HyenaDNA (Nguyen et al., 2023) and Caduceus (Schiff et al., 2024) at up to 450 kb. Of its five tasks, only eQTLP involves a variant. Its model interface embeds the reference and alternate sequences, averages each, concatenates them, and classifies. A model can therefore succeed without distinguishing the alleles at all, if *where* the variant sits predicts the label.

We show that this is what happens. The published split rewards distance and gene identity. Once both are removed, three architectures score the same with and without the alternate allele. We contribute: (1) an audit of the published protocol; (2) **eQTLP-v2**, a distance-matched, chromosome-held-out version with a published floor; (3) a **ref-copy ablation** that any variant benchmark can report; (4) reproducibility findings on the released code; and (5) a matched, evaluation-only ClinVar regulatory set.

## 2 The published protocol rewards distance and memorisation

Positives are SuSiE (Wang et al., 2020) fine-mapped GTEx (The GTEx Consortium, 2020) eQTLs in 9 tissues. Negatives are other variant–gene pairs within 450 kb. Each input spans the variant and the TSS and is right-padded with N to 450 kb. The split is a per-tissue random 8:1:1.

- **Distance beats every model.** Positives lie a median 8–33 kb from the TSS, negatives 88–145 kb. Scoring by  $-\text{distance}$  ( $|\text{variant} - \text{TSS}|$  from the released coordinates, which is what the padded input encodes) gives a test AUROC of 0.688–0.828 per tissue, with a mean of 0.751 (Table 2). That beats the expert model in 7 of 9 tissues and every FM in every tissue. The padding also hands the distance to the model, since the number of N tokens encodes it. Scoring by the released `distance_to_tss` column instead — which disagrees with the coordinates by over 1 kb for 16% of negatives and no positives — gives 0.663: still above every FM, though not the expert.

**Table 1:** Published eQTL test split, per tissue. **dist** is our zero-training baseline, scored from the released coordinates, with a 95% bootstrap CI; **expert** (Enformer), **hyena** and **cad-ph** are the benchmark’s published numbers. Distance wins in every tissue, the CIs are 0.10–0.19 wide, and every test gene appears in training. Full columns in Appendix A.

| Tissue                          | $n_+$ | $n_-$ | dist  | 95% CI       | expert | hyena | cad-ph | gene in train |
|---------------------------------|-------|-------|-------|--------------|--------|-------|--------|---------------|
| Adipose_Subcutaneous            | 16    | 316   | 0.715 | [0.55, 0.87] | 0.736  | 0.513 | 0.541  | 100%          |
| Artery_Tibial                   | 27    | 542   | 0.744 | [0.64, 0.84] | 0.741  | 0.479 | 0.547  | 100%          |
| Cells_Cultured_fibroblasts      | 11    | 350   | 0.688 | [0.54, 0.82] | 0.639  | 0.584 | 0.597  | 100%          |
| Muscle_Skeletal                 | 16    | 397   | 0.798 | [0.68, 0.90] | 0.621  | 0.487 | 0.538  | 100%          |
| Nerve_Tibial                    | 34    | 609   | 0.818 | [0.76, 0.88] | 0.683  | 0.511 | 0.588  | 100%          |
| Skin_Not_Sun_Exposed_Suprapubic | 17    | 456   | 0.828 | [0.73, 0.91] | 0.710  | 0.471 | 0.586  | 100%          |
| Skin_Sun_Exposed_Lower_leg      | 26    | 607   | 0.694 | [0.60, 0.78] | 0.700  | 0.544 | 0.574  | 100%          |
| Thyroid                         | 26    | 524   | 0.699 | [0.58, 0.81] | 0.612  | 0.529 | 0.527  | 100%          |
| Whole_Blood                     | 17    | 305   | 0.776 | [0.66, 0.88] | 0.689  | 0.512 | 0.594  | 100%          |

**Table 2:** Published test AUROC (Table 7 of the benchmark paper), mean over 9 tissues, next to our zero-training baseline. Per-tissue values with CIs are in Appendix A.

| Distance only | Expert (Enformer) | CNN   | HyenaDNA | Caduceus-Ph | Caduceus-PS |
|---------------|-------------------|-------|----------|-------------|-------------|
| <b>0.751</b>  | 0.681             | 0.528 | 0.514    | 0.566       | 0.538       |

- **The test sets cannot rank models.** They hold 11–34 positives per tissue, giving AUROC CI half-widths of  $\pm 0.10$ –0.18 (Hanley and McNeil, 1982), so no published FM result is distinguishable from 0.5. Nor is the split stratified: positive rates are 9–21% in train, 3–5% in test.
- **Train and test overlap.** Every test gene appears in training, and 82–92% of test pairs have a training variant within 100 kb.

Table 1 gives the per-tissue picture. The distance baseline is not winning on average while losing in places: it beats every foundation model in every one of the nine tissues, and the expert model in seven. The confidence intervals are the second half of the problem. With 11–34 positives a test AUROC carries a 95% CI roughly 0.10–0.19 wide, so the published FM numbers (0.47–0.60 across tissues) are not separable from each other, from 0.5, or from the expert model. A benchmark cannot rank models at this sample size whatever the models do.

### 3 eQTL-v2

We pool the tissues into 25,600 unique pairs and drop 111 with conflicting labels. Each positive is matched to one negative on log distance to the TSS, on the same side of it, which gives 2,905 pairs and drops the distance-only AUROC to 0.495. Folds are 5-fold with whole chromosomes held out, giving 570–665 test positives per fold. We also drop the published label-derived mask, which covers the positions of the positive eQTLs.

Matching on distance removes real regulatory signal along with the shortcut, since enhancer–promoter distance genuinely predicts effect. We match anyway, because the task’s stated aim is variant-effect prediction from sequence, and because the published input encodes distance in its padding, so a model can exploit it without reading any sequence. Unmatched results are in Appendix B.

On this configuration, gradient-boosted trees on local composition (GC, CpG, repeats and dinucleotides in windows up to 5 kb) score  **$0.599 \pm 0.031$** . Adding allele features changes this by at most 0.01. On the random split, a gene’s training-set positive rate alone scores 0.70. That is the memorisation the published split allows.

Table 3 separates the two shortcuts by turning each off independently. Distance scores 0.790 whatever the split, because it does not care which genes are held out — and 0.500 once negatives are matched on it. The gene prior behaves in the opposite way: 0.703 when genes recur across the split and 0.387 under chromosome holdout, which is below chance because a gene’s training rate actively misleads on unseen chromosomes. The two are independent, which is why removing only one leaves the task solvable without reading sequence.

The table also quantifies a third problem. The published label mask marks the positions of the positive eQTLs, so simply counting masked bases scores 0.780 on the unmatched task, above every published model.

**Table 3:** What each split lets a simple feature exploit, on eQTL-v2. **all** keeps the original distance imbalance, **matched** removes it; **random** lets genes recur across the split, **chrom** does not. Distance falls 0.79→0.50 under matching, and the gene prior falls 0.70→0.44 under chromosome holdout: those two columns are the shortcuts. The label-mask count collapses from 0.78 to 0.44, confirming it is a label leak. Composition on matched/chrom is the floor; the 0.599 in the text is its mean over 7 matchings  $\times$  10 partitions, while this cell is the single seed-0 matching.

| Feature                         | all / random | all / chrom | matched / random | matched / chrom |
|---------------------------------|--------------|-------------|------------------|-----------------|
| distance to TSS                 | 0.790        | 0.790       | 0.500            | 0.500           |
| gene’s train positive rate      | 0.703        | 0.387       | 0.706            | 0.439           |
| distance + label-mask count     | 0.810        | 0.780       | 0.529            | 0.436           |
| local composition               | 0.744        | 0.639       | 0.665            | 0.588           |
| distance + composition          | 0.833        | 0.790       | 0.670            | 0.582           |
| distance + composition + allele | 0.834        | 0.790       | 0.670            | 0.590           |

It falls to 0.436 once the positives are distance-matched, confirming it is a label leak rather than signal. We drop the mask in eQTL-v2 for this reason.

## 4 No model uses the allele

**Protocol.** The experiment is a controlled substitution. We score every variant twice with the same already-fitted model: once with its real alternate allele, and once with that allele overwritten by the reference base, so the two inputs differ at exactly one position and in nothing else. A model that reads the variant must give the two different scores. A model that is really scoring the variant’s location will give the same score both times.

Concretely, each model converts a sequence into a vector: it produces one internal state per base, and those states are averaged across all positions into a single summary vector  $e$ . The benchmark’s own interface then concatenates the reference and alternate summaries and classifies the pair. We report three quantities from it:

- **refalt** — the test AUROC on the real pair  $[e_{\text{ref}}, e_{\text{alt}}]$ ;
- **ref\_copy** — *the same fitted classifier*, nothing refitted, applied to  $[e_{\text{ref}}, e_{\text{ref}}]$ , so the alternate allele is gone;
- **diff\_only** — a classifier trained only on the difference  $e_{\text{alt}} - e_{\text{ref}}$ , which asks whether the allele’s effect is recoverable at all.

The models are the benchmark’s **CNN** encoder, trained end to end with its collapsing logit ReLU repaired (§5); and **HyenaDNA-medium-450k** (Nguyen et al., 2023) (which reads sequence in one direction only, 6.55 M parameters) and **Caduceus-Ph-131k** (Schiff et al., 2024) (which reads both directions, 7.73 M). The latter two are used *frozen*: their pretrained weights are never updated, and we fit only a logistic regression on top of the summary vectors they produce (Pedregosa et al., 2011) — a *linear probe*, in the representation-learning sense of that word rather than the molecular one. This is standard practice for asking what information a pretrained representation already contains, and we report below what happens when we instead fine-tune the whole network. Inputs are unpadded, and for each model we try both the published sequence orientation and its reverse complement.

- **The allele changes nothing** (Figure 1a). Replacing it with the reference changes AUROC by at most 0.0017 for the trained CNN (15 runs) and 0.0006 for the FM probes, across probe strengths ( $C = 0.001$ –10), heads (linear, MLP), orientations, and 13 fold partitions (chromosome, gene and random). The paired 95% CI is  $[-0.0002, 0.0002]$ .
- **Only a weak trace of the allele is there to find, and pooling is not what hides it.** A linear **diff\_only** probe scores at chance (0.49–0.51). An MLP on the difference scores at or above a 20-permutation label null ( $p = 0.048$ –0.095; Caduceus 0.53–0.55, HyenaDNA 0.52–0.53) but stays below the floor, and an MLP head on  $[e_{\text{ref}}, e_{\text{alt}}]$ , like the released Caduceus head, still has  $\Delta = 0.0000$ .

Averaging over the whole sequence discards position, so we next ask whether the allele survives somewhere that average cannot reach. We repeat the search at every layer and over four *summary windows* — the set of positions averaged together before classifying: the whole sequence,  $\pm 1$  kb around the variant,  $\pm 64$  bp, and the single variant position itself. Each is scored against a control in which the variant is replaced by a base that did **not** occur, which absorbs anything attributable to local sequence context rather than to the allele. For HyenaDNA the control matches the signal everywhere (mean  $-0.006$  over 36 cells), so what

**Table 4:** eQTLP-v2, matched pairs, chromosome holdout. Mean per-fold test AUROC  $\pm$  SD.

| Model                                   | refalt           | ref_copy         | $\Delta$ | diff_only        |
|-----------------------------------------|------------------|------------------|----------|------------------|
| CNN (trained, 3 seeds $\times$ 5 folds) | 0.517 $\pm$ .040 | 0.517            | -0.0002  | —                |
| HyenaDNA (frozen)                       | 0.556 $\pm$ .031 | 0.556 $\pm$ .030 | -0.0001  | 0.495 $\pm$ .021 |
| Caduceus-Ph (frozen)                    | 0.564 $\pm$ .021 | 0.564 $\pm$ .021 | 0.0000   | 0.507 $\pm$ .016 |
| Local-composition floor                 | 0.599 $\pm$ .031 |                  |          |                  |

looked like allele information was context. For Caduceus one window survives its control — the final layer at  $\pm 1$  kb, scoring 0.575–0.580 against 0.545–0.553, on all three fold schemes. So a variant-centred summary does recover allele signal that whole-sequence averaging loses. It is still below the floor, and the swap still changes nothing even there (max  $|\Delta|$  0.017 over 104 cells, always favouring **ref\_copy**).

- **All three models score below the hand-feature floor** (Figure 1a), in 10 of 10 random chromosome partitions. Only on the random split, which allows locus memorisation, do the FMs pass it (0.69–0.70 vs 0.67).

Figure 2 shows that search in full. Each cell is one layer read out one way, scored by how much a nonlinear probe on the allele difference beats the same probe on a control in which the variant is replaced by a base that never occurred. The control is what makes the comparison informative: it absorbs everything the probe could learn from local sequence context, so what remains is attributable to the allele itself. Three things are visible. HyenaDNA has no cell that clears its control, which is why we describe its trace as context rather than allele. Caduceus is warm in a band — the  $\pm 1$  kb and  $\pm 64$  bp readouts, across most of the depth — but almost all of it is within noise of the control. And exactly one cell stands clear, the final layer at  $\pm 1$  kb, which replicates on all three fold schemes. That cell is the strongest allele-specific signal anywhere in either model, and at 0.575–0.580 it is still below the 0.599 hand-feature floor.

**The SNP barely registers** (Figure 1b). The median relative shift  $\|e_{\text{alt}} - e_{\text{ref}}\|/\|e_{\text{ref}}\|$  is  $1.5\text{--}2.0 \times 10^{-4}$  for HyenaDNA and  $0.7 \times 10^{-4}$  for Caduceus. eQTLP-v2 is 92.3% single-nucleotide substitutions and 7.7% short indels or MNVs; restricting to the substitutions alone leaves both medians unchanged ( $1.52 \times 10^{-4}$  and  $0.674 \times 10^{-4}$ ), so this is a per-variant effect and not an artifact of the indels. The CNN’s max-pooled embedding changes in only 0.4–1.7% of pairs. Two explanations for this fail.

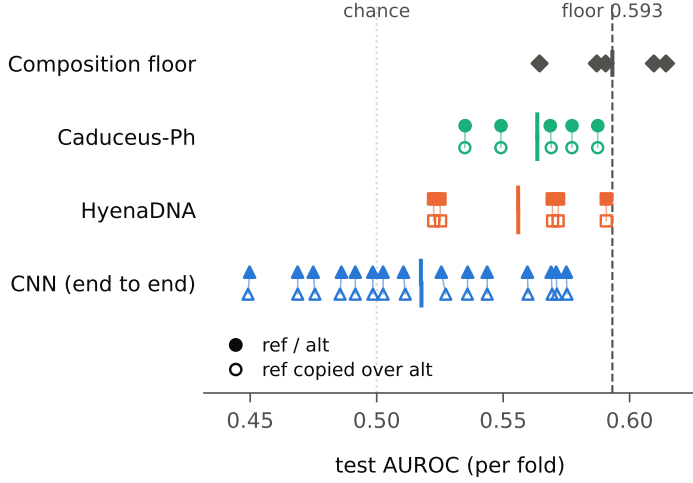
**Orientation is not the cause.** HyenaDNA is causal, and the published orientation places the variant roughly 500 bp from the end of the input, so in 71–86% of pairs only a few hundred positions can attend to it. This is a real defect of the protocol, and it is the obvious suspect. It is not the explanation. Embedding the reverse complement instead, so that the variant comes first and every subsequent position can see it, moves AUROC by less than 0.01 and leaves the **ref\_copy** gap in the fourth decimal. Caduceus is bidirectional by construction and behaves identically to HyenaDNA throughout. A null that survives both reading orders, in both a causal and a bidirectional architecture, is not an artifact of where the variant sits in the window.

**Numerical precision is not the cause.** A relative shift of  $10^{-4}$  in bf16 invites the objection that the allele is being rounded away. We re-embedded 400 pairs in full fp32: the shifts reproduce, the difference vectors agree with their bf16 counterparts at cosine 0.89–0.95, and **diff\_only** stays at chance. The signal that a probe cannot find in bf16 is equally absent at full precision, so the model is not computing an allele effect that the numerics discard.

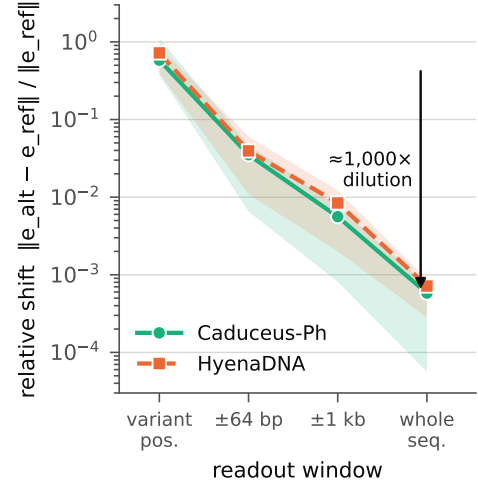
**What remains is pooling.** The substitution does move the representation, and locally it moves it a great deal: the variant-position hidden state changes by 37–112%, the  $\pm 64$  bp mean by 0.7–6.1%, and the whole-sequence mean by only 0.006–0.09%. Taking medians across layers, what reaches the classifier is about  $1,000\times$  smaller than the change at the variant itself. Averaging across tens of kilobases is simply the wrong way to summarise a one-base question, and it is the summary the benchmark specifies. This is why we recommend summarising a window around the variant in §9 — but note the limit of that recommendation: Figure 2 shows that even the best local window recovers only a signal that remains below the hand-feature floor.

**Fine-tuning does not change it.** The probes above are frozen, whereas the published numbers come from fine-tuning, so we fine-tuned HyenaDNA-medium-450k end to end on eQTLP-v2 — backbone and head, through the benchmark’s own mean-pooled readout — for 2 epochs on each of 3 chromosome folds

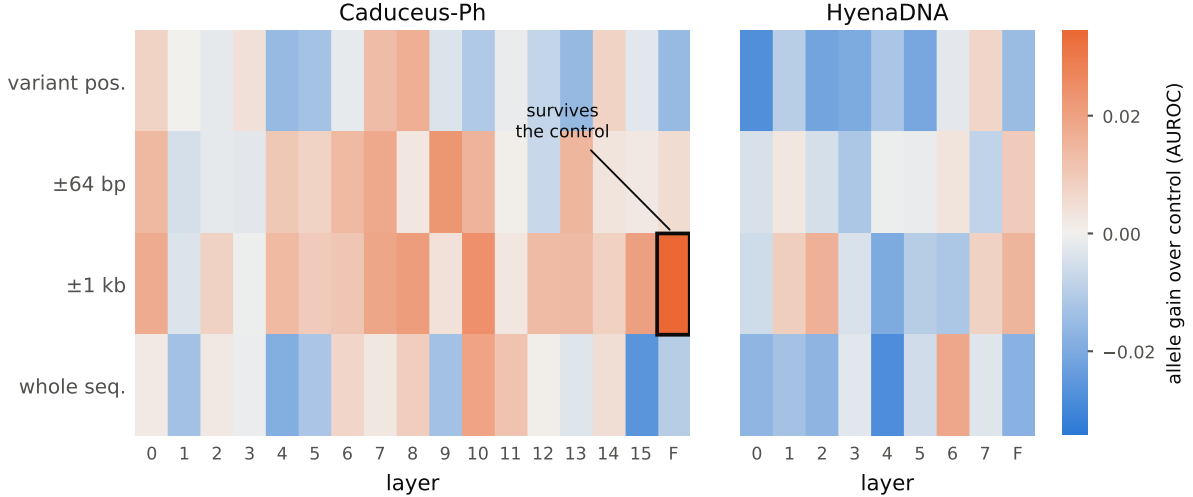
### A No model uses the allele



### B Pooling dilutes the substitution



**Figure 1:** (a) eQTL-v2, matched pairs, chromosome holdout: per-fold test AUROC for each model, scored normally (**refalt**, filled) and with the reference copied over the alternate allele (**ref\_copy**, open). Each fold’s pair is joined by a line, and every line is vertical: the swap moves no score. Points are 5 folds ( $\times 3$  seeds for the CNN); the bar is the mean. The dashed line is the floor on these same folds ( $0.593$ ); §3 quotes  $0.599 \pm 0.031$ , its mean over 7 matchings  $\times 10$  partitions. No model reaches it. (b) Relative shift  $\|e_{\text{alt}} - e_{\text{ref}}\| / \|e_{\text{ref}}\|$  under four summary windows, from the variant position out to a whole-sequence average; markers are the median over layers, bands the min-max.



**Figure 2:** Allele signal by layer and summary window, for both frozen models. Colour is the gain of a nonlinear probe on  $e_{\text{alt}} - e_{\text{ref}}$  over the same probe on a random-allele control, so 0 means the probe was reading local context rather than the variant. **F** is the pooled final-layer state. Warm cells concentrate in the local windows and only one (Caduceus, final layer,  $\pm 1$  kb) clears its control by a margin; it is still below the floor, and the **ref\_copy** swap changes nothing there.

(33 GPU-hours; inputs capped at 131 kb, which 88% of pairs fit entirely).

Training works. Loss falls from 0.730 to 0.698, and **refalt** rises across epochs on 2 of the 3 folds (fold 0:  $0.543 \rightarrow 0.563$ ; fold 1:  $0.565 \rightarrow 0.572$ ; fold 2 regresses,  $0.566 \rightarrow 0.529$ ). The final-epoch mean is  $0.555 \pm 0.023$ , statistically indistinguishable from the frozen probe’s  $0.556 \pm 0.031$  and still below the  $0.599$  floor: updating every weight in the backbone buys nothing over a logistic regression on its frozen features.

**Table 5:** End-to-end fine-tuning of HyenaDNA-medium-450k on eQTLp-v2, through the benchmark’s own readout. Training works — loss falls and **refalt** rises on 2 of 3 folds — and lands at the frozen probe’s accuracy, but the allele gap never leaves the fourth decimal.

| Fold | Epoch | refalt | ref_copy | $\Delta$ | train loss | $n$  | $h$ |
|------|-------|--------|----------|----------|------------|------|-----|
| 0    | 1     | 0.5430 | 0.5429   | +0.00008 | 0.7295     | 1212 | 4.6 |
| 0    | 2     | 0.5632 | 0.5633   | -0.00004 | 0.7064     | 1212 | 6.5 |
| 1    | 1     | 0.5651 | 0.5652   | -0.00001 | 0.7126     | 953  | 4.6 |
| 1    | 2     | 0.5722 | 0.5721   | +0.00016 | 0.7020     | 953  | 5.4 |
| 2    | 1     | 0.5663 | 0.5663   | +0.00005 | 0.7126     | 1083 | 5.4 |
| 2    | 2     | 0.5293 | 0.5293   | +0.00002 | 0.6983     | 1083 | 5.5 |

The allele gap does not move. Across all six epochs the largest  $|\Delta|$  is **0.00016**, with a mean of +0.00005 — the same fourth-decimal nothing the frozen probes gave. The model measurably improves at the task while learning nothing about which allele is present. This is the strongest form of the result we can offer: the null is not an artifact of freezing the backbone.

**Scope and limitations.** The fine-tuning run is one model, one seed, 2 epochs per fold at 131 kb rather than the published 450 kb, so it bounds what this much training achieves, not what any training could. We evaluate two models and both are small (6.55 M and 7.73 M parameters); larger long-context models such as Evo 2 (Brix et al., 2026) are beyond our compute. Our claim is therefore about what these architectures and this readout do on this task, not about model scale.

## 5 The released code does not run as shipped

We report defects found by reading the released code and running it on real records. We do not claim to have re-run the published experiments and obtained different numbers, which would be a different kind of result: with these defects the released code cannot produce the published FM numbers at all, so those numbers must come from a version that is not the one released. We raise this as a reproducibility problem for the artifact, not as a conclusion about what the original authors ran.

- **The FM eQTL loaders produce empty inputs.** Both the HyenaDNA and Caduceus loaders pad each input to 450,000 positions, then slice it with [2028500:-2028500], leaving an empty array.
- **The token ids are wrong.** Without the slice, `argmax` over the one-hot gives ids 0–3, which are special tokens in the pretrained vocabularies.
- **The configs don’t match the checkpoints.** They specify 2 layers, while the checkpoints have 8 and 16, and the pretrained-weights path is unset.
- **The CNN’s training often collapses.** Its eQTLp head ends in a ReLU on the logits, so both logits can clamp to zero: the prediction is constant and every weight gets zero gradient. On the *published* split, 2 of 10 initialisations are dead before the first step, and after one epoch 5–6 of 10 seeds output a constant score (AUROC exactly 0.5). The script sets no seed and evaluates the final-epoch model, not the best. The CNN is fed only the reference.

## 6 Clinical evaluation set

ClinVar (Landrum et al., 2025) pathogenic or likely-pathogenic regulatory SNVs with at least 1 star, annotated against GENCODE (The GENCODE Consortium, 2025), must be non-coding, over 50 bp from a splice site, off every CDS, and outside small structural RNAs; 652 map to a gene TSS within 450 kb. Each is matched to a benign SNV passing the same filters, on distance to the TSS and, where possible, the same gene and location class (555 of 652 pairs). No model is fitted on clinical labels, and the embedding shift, the transferred eQTLp-v2 probe and that probe’s allele-only component are all at chance (AUROC 0.489–0.507 overall, 0.49–0.52 on the 285 pairs with a 2-star-or-better pathogenic). The only stratum above chance is 5’UTR/promoter (0.53–0.57), where matching left pathogenic variants closer to the TSS (median 170 vs 450 bp) and distance alone scores 0.61. The same null therefore appears under pathogenicity labels; the two targets differ, so this corroborates the eQTL result rather than generalising it. The set is evaluation-only because too few regulatory positives survive the filters to train on (409 at 2 stars).

The set is small and concentrated: a handful of well-studied genes contribute many pairs each. Appendix F repeats the evaluation after dropping the three most frequent genes, after dropping the ten most frequent, and after keeping one pair per gene. Every score stays at chance, so the null is not an artifact of which genes ClinVar happens to have annotated densely.

## 7 Relation to prior work

Two long-range DNA benchmarks are adjacent to ours: the Genomics Long-Range Benchmark (Kao et al., 2024), with an OMIM pathogenic non-coding variant task, and BEND (Marin et al., 2024), on reference-sequence element annotation. Neither reports an allele ablation, the diagnostic we argue for: without it, a variant task can be passed by locating the variant rather than reading it. Our result also extends to a benchmark setting the finding that sequence-to-expression models predict individual expression variation poorly (Sasse et al., 2023; Huang et al., 2023). The failure mode is shortcut learning (Geirhos et al., 2020): the protocol, not the architecture, decides what a model must learn.

**Distance is a known baseline, not a surprise.** That genomic distance predicts enhancer–gene links is long established: the activity-by-contact model combines enhancer activity with contact frequency, which falls off steeply with distance, and outperforms many sequence-based predictors of which enhancer regulates which gene (Fulco et al., 2019). The finding here is therefore not that distance is informative — it is that a benchmark advertised as measuring variant-effect prediction from sequence can be won by distance alone, and that none of the evaluated models beats it. A field that already knows distance is strong should be reporting it as a baseline.

**Perturbation data avoids the fine-mapping confound.** Our labels are statistically fine-mapped eQTLs, which inherit the confounding structure of linkage disequilibrium: the credible set marks variants correlated with an expression effect, not necessarily the causal one. CRISPR interference screens that perturb candidate elements directly and read out expression (Gasperini et al., 2019; Schraivogel et al., 2020) give element–gene links that do not depend on fine-mapping assumptions, and they are a better foundation for a variant benchmark than association statistics are — though they test elements rather than single bases, so they do not by themselves solve the allele problem we identify.

**Model scale.** We evaluate two small models. Larger genomic foundation models exist — Nucleotide Transformer (Dalla-Torre et al., 2025) at up to 2.5 B parameters, Evo 2 (Brix et al., 2026) at 7 B and 40 B — and we have not tested whether scale changes the picture. Two considerations cut against assuming it does. The pooled readout that dilutes the substitution  $\sim 1,000\times$  is a property of the interface, not the backbone, so a larger backbone read out the same way inherits the same dilution. And pretraining on a single reference genome provides no objective that rewards distinguishing alleles at a locus, whatever the parameter count. Neither argument is a substitute for running the experiment.

## 8 Discussion

**What this does and does not show.** It shows that DNALongBench’s eQTL task, as published, can be passed without reading the alternate allele, and that three architectures do exactly that. It does not show that long-context DNA models are incapable of variant-effect prediction. The distinction matters because the two conclusions call for different responses: the first is a protocol that needs fixing, the second would be an architectural verdict we have no basis to issue. Our evidence is two frozen models of 6.55 M and 7.73 M parameters plus one CNN trained end to end, on one task, under the benchmark’s own readout. A larger model, a variant-aware readout, or a training signal that rewards allele discrimination could all change the result — and the fact that none of them has been tested is a gap in the field, not a settled question.

**Why a protocol can hide this.** The failure is legible only because we looked for it. A benchmark reports one number per model, and 0.566 for Caduceus-Ph is unremarkable: neither so high as to invite an audit nor so low as to suggest the task is broken. Nothing in that number reveals that a zero-training distance baseline scores 0.751 on the same split, that the test set holds 27 positives, or that replacing the alternate allele with the reference leaves the score unchanged. Each of those is a cheap check. None is standard.

**The ablation generalises past this task.** The `ref_copy` construction needs no retraining and no new data: take the fitted model, feed it an input in which the quantity under test has been neutralised, and score

again. It applies wherever a benchmark claims a model responds to a specific perturbation — a variant, an edit, a knockout, a designed mutation. We think a variant benchmark that does not report it has not established that its models read variants, and that the burden should sit with the benchmark rather than with whoever audits it afterwards.

**On the value of floors.** The hand-feature floor did more diagnostic work here than any model comparison. A floor of 0.599 from GC content, CpG density, repeats and dinucleotide frequencies is not a strong result; what makes it useful is that three deep models fail to reach it. Model-versus-model tables cannot produce that observation, because every entry can be mediocre together. Publishing a floor costs a few CPU-hours and converts an unremarkable leaderboard into a falsifiable claim.

## 9 Recommendations

Variant benchmarks for long-context models should match negatives on distance to the TSS and, where possible, gene; hold out chromosomes; size test sets for  $\pm 0.05$  CIs; publish a hand-feature floor; report a **ref-copy ablation** beside every variant score; and read the variant out *locally* rather than as a whole-input mean.

On DNALongBench’s eQTL task, current models learn where a variant is, not what it is.

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## Appendix

Appendices A–F are submitted as Supplementary Material (`appendix.pdf`).